

EXPERIMENT (7)

Ex. No.: 7

Date:

AIM

To retrieve records from the Hostel Management System tables using GROUP BY, HAVING and ORDER BY clauses.

DESCRIPTION

GROUP BY groups rows having the same values. HAVING filters grouped records. ORDER BY sorts records in ascending or descending order.

SYNTAX

GROUP BY - HAVING

```
SELECT column_name, aggregate_function(column_name)
FROM table_name
GROUP BY column_name
HAVING condition;
```

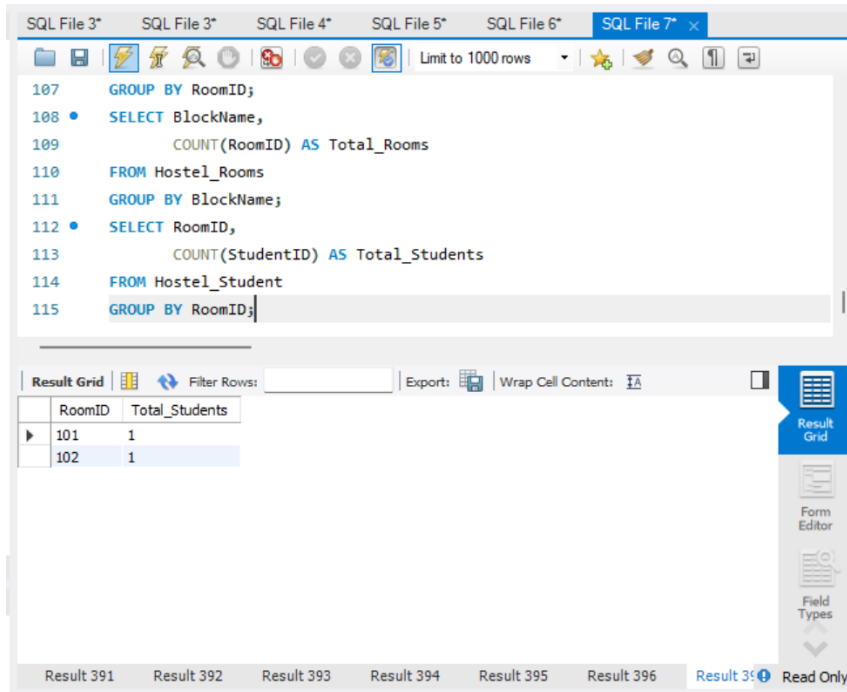
ORDER BY

```
SELECT column1, column2
FROM table_name
ORDER BY column_name ASC|DESC;
```

Questions

1. Display the number of students staying in each room.

```
SELECT RoomID,  
       COUNT(StudentID) AS Total_Students  
FROM Hostel_Student  
GROUP BY RoomID;
```



The screenshot shows a SQL IDE interface with a query editor and a results grid. The query editor contains the following SQL code:

```
107 GROUP BY RoomID;  
108 • SELECT BlockName,  
109       COUNT(RoomID) AS Total_Rooms  
110 FROM Hostel_Rooms  
111 GROUP BY BlockName;  
112 • SELECT RoomID,  
113       COUNT(StudentID) AS Total_Students  
114 FROM Hostel_Student  
115 GROUP BY RoomID;
```

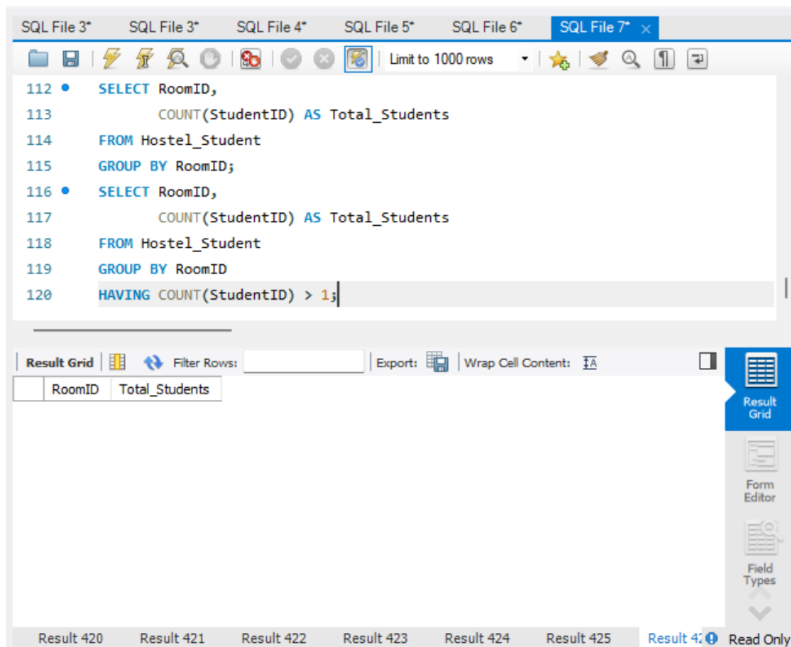
The results grid displays the following data:

RoomID	Total_Students
101	1
102	1

The IDE interface includes a toolbar with various icons, a status bar at the bottom showing "Result 391" through "Result 396", and a "Read Only" indicator.

2. Display only the rooms having more than one student.

```
SELECT RoomID,  
       COUNT(StudentID) AS Total_Students  
FROM Hostel_Student  
GROUP BY RoomID  
HAVING COUNT(StudentID) > 1;
```

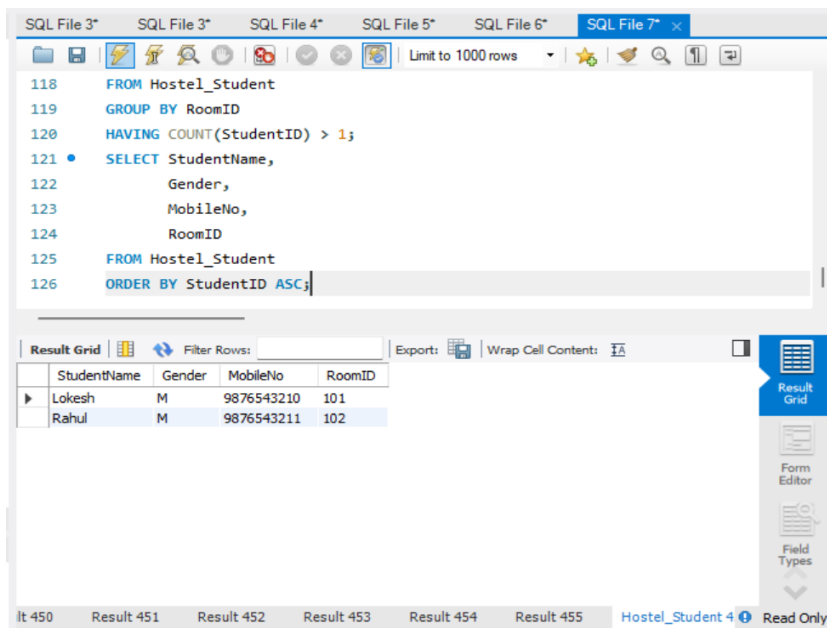


3. Display Student Name, Gender, Mobile Number and RoomID in ascending order of StudentID.

```

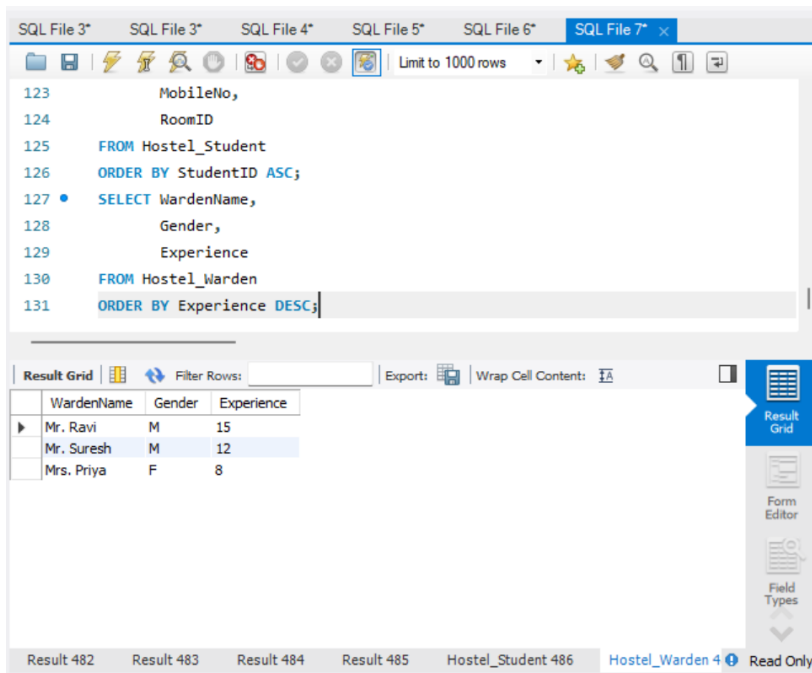
SELECT StudentName,
       Gender,
       MobileNo,
       RoomID
FROM Hostel_Student
ORDER BY StudentID ASC;

```



4. Display wardens in descending order of experience.

```
SELECT WardenName,  
       Gender,  
       Experience  
FROM Hostel_Warden  
ORDER BY Experience DESC;
```



The screenshot shows a SQL IDE interface with multiple tabs labeled 'SQL File 3*' through 'SQL File 7*'. The active tab is 'SQL File 7*'. The query editor displays the following SQL code:

```
123      MobileNo,  
124      RoomID  
125  FROM Hostel_Student  
126  ORDER BY StudentID ASC;  
127  • SELECT WardenName,  
128          Gender,  
129          Experience  
130  FROM Hostel_Warden  
131  ORDER BY Experience DESC;
```

Below the query editor, the 'Result Grid' is visible, showing the results of the query. The grid has columns for WardenName, Gender, and Experience. The results are as follows:

WardenName	Gender	Experience
Mr. Ravi	M	15
Mr. Suresh	M	12
Mrs. Priya	F	8

The interface also includes a 'Filter Rows' section, an 'Export' button, and a 'Wrap Cell Content' option. The bottom status bar shows 'Result 482', 'Result 483', 'Result 484', 'Result 485', 'Hostel_Student 486', 'Hostel_Warden 487', and 'Read Only'.

RESULT

Thus, the records were successfully retrieved from the Hostel Management System tables using GROUP BY, HAVING and ORDER BY clauses.